

# Mobile Sales Data Analysis

## 1. Project Overview

This project analyzes mobile phone sales performance using transactional data covering 3,835 purchases recorded between October 2021 and October 2024 across 19 Indian cities. The goal is to uncover insights into brand performance, top-selling models, payment preferences, regional demand, and customer ratings to guide strategic business decisions.

## 2. Dataset Summary

- Rows: 3,835
- Columns: 14 (raw) → 17 after feature engineering

**Key Features:**

- Transaction details (Transaction ID, Day/Month/Year, Day Name, Purchase Date)
  - Product details (Brand, Mobile Model, Units Sold, Price per Unit, Total Sale Amount)
  - Customer details (Customer Name, Customer Age, Age Group, City)
  - Shopping behaviour (Payment Method, Customer Ratings)
- Missing Data: 0 null values across all columns
  - Data Quality Issue Found: 23 rows used abbreviated weekday labels (e.g. 'Wed', 'Fri') inconsistent with the full-name convention used elsewhere; these were standardized during cleaning.

## 3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

**Data Loading:** Imported the dataset using pandas.

**Initial Exploration:** Used df.info() and .describe() for summary statistics.

| Metric | Units Sold | Price/Unit (USD) | Customer Age | Rating | Total Sale (USD) |
|--------|------------|------------------|--------------|--------|------------------|
| count  | 3,835      | 3,835            | 3,835        | 3,835  | 3,835            |
| mean   | 5.0        | 40,114.04        | 38.1         | 3.69   | 200,574.96       |
| min    | 1          | 10,011.11        | 18           | 1      | 10,115.91        |
| 25%    | 3          | 25,617.41        | 27.5         | 3      | 83,930.94        |
| 50%    | 5          | 40,136.49        | 38           | 4      | 163,855.84       |
| 75%    | 7          | 54,973.48        | 49           | 5      | 289,246.94       |
| max    | 9          | 69,984.48        | 59           | 5      | 628,391.79       |

**Missing Data Handling:** Checked for null values; none were found in this dataset.

**Column Standardization:** Renamed columns to snake\_case for readability and documentation.

**Feature Engineering:**

- Created **purchase\_date** from the Day, Month, and Year columns.
- Created **total\_sale\_amount** as units\_sold × price\_per\_unit.
- Created **age\_group** column by binning customer ages into quartiles (Young Adult, Adult, Middle-aged, Senior).

**Data Consistency Check:** Verified that total\_sale\_amount matches units\_sold × price\_per\_unit for 100% of rows, confirmed 0 duplicate rows, and standardized the abbreviated day\_name values noted above.

**Database Integration:** Connected the Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

## 4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

**1. Revenue & Units by Brand** – Compared total revenue and units sold across all five brands.

| Brand   | Total Revenue (USD) | Total Units |
|---------|---------------------|-------------|
| Apple   | 161,615,730         | 3,932       |
| Samsung | 160,038,054         | 3,923       |
| OnePlus | 153,719,439         | 3,830       |
| Vivo    | 150,078,428         | 3,801       |
| Xiaomi  | 143,753,337         | 3,664       |

**2. Top 5 Mobile Models by Units Sold**

| Mobile Model   | Total Units Sold |
|----------------|------------------|
| iPhone SE      | 1,430            |
| Vivo Y51       | 1,429            |
| OnePlus Nord   | 1,409            |
| Galaxy Note 20 | 1,382            |
| Galaxy S21     | 1,305            |

**3. Average Customer Rating by Payment Method**

| Payment Method | Avg. Rating |
|----------------|-------------|
| Debit Card     | 3.73        |
| Cash           | 3.68        |
| Credit Card    | 3.68        |
| UPI            | 3.68        |

4. Top 5 Cities by Revenue

| City    | Revenue (USD) |
|---------|---------------|
| Delhi   | 203,883,476   |
| Mumbai  | 127,191,280   |
| Ranchi  | 31,018,767    |
| Chennai | 30,759,534    |
| Rajkot  | 27,326,701    |

5. Sales Trend by Year and Payment Method

| Year | Payment Method | Orders | Revenue (USD) |
|------|----------------|--------|---------------|
| 2021 | Credit Card    | 80     | 15,749,248    |
|      | Cash           | 67     | 15,537,644    |
|      | UPI            | 76     | 14,717,112    |
|      | Debit Card     | 73     | 12,811,517    |
| 2022 | Debit Card     | 313    | 67,849,075    |
|      | Cash           | 329    | 65,513,798    |
|      | UPI            | 315    | 65,383,983    |
|      | Credit Card    | 324    | 63,242,984    |
| 2023 | UPI            | 348    | 68,137,721    |
|      | Debit Card     | 325    | 64,487,898    |
|      | Cash           | 307    | 60,674,350    |
|      | Credit Card    | 305    | 60,014,251    |
| 2024 | UPI            | 272    | 53,434,945    |
|      | Debit Card     | 237    | 50,590,292    |
|      | Credit Card    | 238    | 47,691,997    |
|      | Cash           | 226    | 43,368,174    |

6. Revenue by Age Group

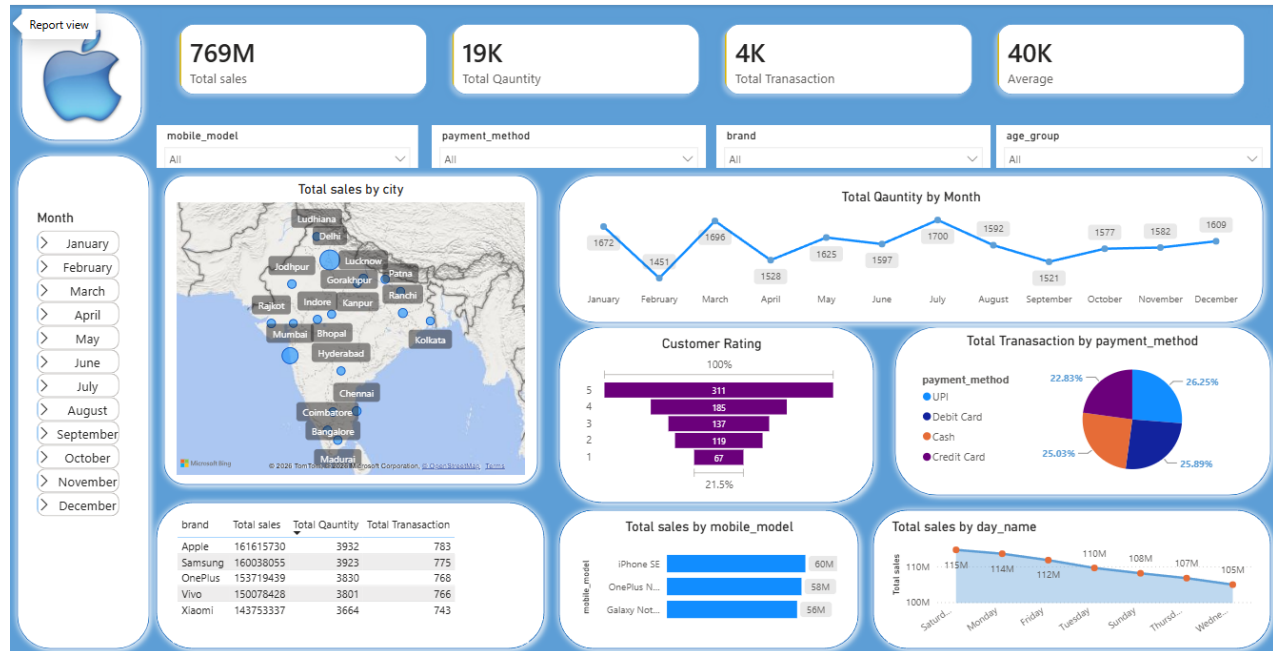
| Age Group   | Total Revenue (USD) |
|-------------|---------------------|
| Adult       | 198,278,366         |
| Middle-aged | 196,029,679         |
| Young Adult | 191,556,466         |

Senior

183,340,477

## 5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



## 6. Business Recommendations

- **Prioritize Apple and Samsung** – They generate the highest revenue and unit volume; ensure strong inventory and marketing focus.
- **Promote UPI and Debit Card Payments** – UPI drives the highest order volumes in 2023-2024, and Debit Card users report the highest satisfaction; incentivize these channels.
- **Double Down on Delhi and Mumbai** – These two cities alone contribute over 43% of total revenue; deepen distribution and localized campaigns there.
- **Feature Top Models** – Highlight iPhone SE, Vivo Y51, and OnePlus Nord in campaigns as consistent best-sellers.
- **Balanced Age-Group Targeting** – Revenue is fairly evenly spread across age groups, so broad-based marketing outperforms narrow age targeting.
- **Data Quality Governance** – Standardize categorical fields (e.g. weekday labels) at the point of entry to reduce downstream cleaning effort.